



## Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

### DEPARTMENT OF CSE

#### PROJECT ABSTRACT SUBMISSION

### COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

| TITLE OF THE PROJECT | Smart Canteen Ordering and Management System Using Spring Boot and React |                            |
|----------------------|--------------------------------------------------------------------------|----------------------------|
| S. No.               | University ID                                                            | Name                       |
| 1                    | 2420030609                                                               | Rajendra Phalani           |
| 2                    | 2420030615                                                               | D Mallikarjun              |
| 3                    | 2420030370                                                               | D Praveen                  |
| Name of the Guide    |                                                                          | <b>Dr.Neelima Gogineni</b> |

#### Abstract:

The **Smart Canteen Ordering and Management System Using Spring Boot and React** is a modern web-based application developed to automate and improve the traditional college canteen ordering process. In conventional canteens, students often need to stand in long queues, wait for food preparation, and manually communicate their orders to the canteen staff. These processes can lead to long waiting times, incorrect orders, difficulty in managing peak-hour demand, and challenges in maintaining accurate order records. The proposed system provides a centralized digital platform that connects students and canteen staff and makes the complete ordering and pickup process more efficient.

The system provides a user-friendly interface through **React.js**, where students can register, log in, browse the available food menu, view prices and availability, select food items, add them to a cart, and place orders. The **Spring Boot** backend manages authentication, business logic, REST APIs, order processing, and communication between the frontend and database. **MySQL** is used to securely store information such as users, food items,

categories, orders, order items, prices, and order statuses. Authentication can be implemented using **Spring Security and JWT** to provide secure access to different user roles.

A separate **admin/canteen dashboard** allows authorized staff to manage the complete canteen operation. Staff can add, update, or remove food items, change food availability, view incoming orders, and update order statuses. After a student places an order, it follows a structured workflow such as **Placed → Confirmed → Preparing → Ready → Completed**. When the food is ready, the student receives an in-application notification and can visit the pickup counter. The student can display an **Order ID or QR code**, which the canteen staff verifies through the system before handing over the food. After successful verification, the order is marked as completed, helping prevent duplicate or incorrect order collection.

The system can also provide **order history, sales reports, food availability tracking, low-stock alerts, and basic analytics** to help canteen administrators understand demand and improve daily operations. By replacing manual ordering with a digital workflow, the system reduces queue time, minimizes human errors, improves order tracking, and provides better coordination between students and canteen staff.

Overall, the proposed system offers a **scalable, secure, and user-friendly digital solution for college canteens**. The combination of React, Spring Boot, REST APIs, MySQL, and QR-based pickup verification creates an efficient end-to-end platform covering menu management, online ordering, order preparation, notification, verification, and final pickup. The system can be further enhanced in the future with **online payment integration, real-time notifications, AI-based demand prediction, personalized food recommendations, and advanced sales and inventory analytics**.

**Project Architecture/Project Overview:**

